

Chapter 7: Discrete Fourier Transform

Spectral analysis of discrete time signal is usually and must conveniently perform on a digital signal processor, which may be a general – purpose digital computer or especially digital hardware. To perform spectral analysis on a discrete time signal $x[n]$ we convert the time domain sequence to an equivalent frequency – domain representation. We know that such a representation is given by the Fourier transform $X(j\omega)$ or $X(e^{j\omega})$ of the sequence $x[n]$. However $X(e^{j\omega})$ is a continuous function of frequency and therefore, it is not a computationally convenient representation of the sequence $x[n]$.

Now to overcome these discrepancies we consider the representation of a sequence $x[n]$ by sample of its spectrum $X(e^{j\omega})$. Such a frequency – domain representation leads to discrete Fourier transform (DFT), which is a powerful computational tool for perform frequency analysis of discrete – time signals.

Frequency domain sampling of discrete time signal

Before introducing DFT, we consider the sampling of the Fourier transform of an aperiodic discrete time sequence. Thus we establish the relationship between the sampled Fourier transform and the DFT.

Consider an aperiodic finite – energy signal $x[n]$ with Fourier transform,

$$\text{i. e. } X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \dots \dots \dots (7.1)$$

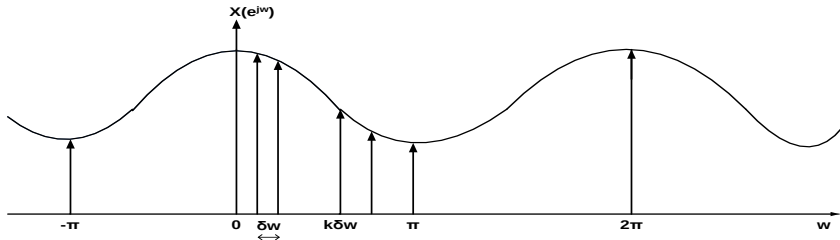


Figure 7.1: Frequency- domain sampling of the Fourier transform

Suppose that we sample $X(e^{j\omega})$ periodically in frequency at a spacing of $\delta\omega$ radians between successive samples. Since $X(e^{j\omega})$ is periodic with period 2π , only sample in the fundamental frequency range are necessary. For convenience we take N equidistant samples in the interval $0 \leq \omega < 2\pi$ with spacing $\delta\omega = \frac{2\pi}{N}$ as shown in figure 7.1.

At $\omega = \frac{2\pi k}{N}$, from equation 7.1, we get,

$$X\left(e^{j\frac{2\pi k}{N}}\right) = \sum_{n=-\infty}^{\infty} x[n]e^{-\frac{j2\pi kn}{N}} \quad k = 0, 1, \dots, N-1$$

This summation can be subdivided into an infinite number of summations, where each sum contains N terms. Thus

$$X\left(e^{j\frac{2\pi k}{N}}\right) = \dots + \sum_{n=-N}^{-1} x[n] e^{-\frac{j2\pi kn}{N}} + \sum_{n=0}^{N-1} x[n] e^{-\frac{j2\pi kn}{N}} + \sum_{n=N}^{2N-1} x[n] e^{-j2\pi kn/N}$$

$$\therefore X\left(e^{j\frac{2\pi k}{N}}\right) = \sum_{l=-\infty}^{\infty} \sum_{n=lN}^{lN+N-1} x[n]e^{-\frac{j2\pi kn}{N}}$$

If we change the index in the inner summation from n to n – lN and interchange the order of the summation, we obtain the result as,

$$X\left(e^{j\frac{2\pi k}{N}}\right) = \sum_{n=0}^{N-1} \left[\sum_{l=-\infty}^{\infty} x[n - lN] \right] e^{-\frac{j2\pi kn}{N}}$$

$$X\left(e^{j\frac{2\pi k}{N}}\right) = \sum_{n=0}^{N-1} x_p[n]e^{-\frac{j2\pi kn}{N}} \dots \dots \dots (7.2)$$

where, $x_p[n] = \sum_{l=-\infty}^{\infty} x[n - lN]$ is obtained by the periodic repetition of x[n] every

N sample and is clearly periodic with fundamental period N

Therefore $x_p[n]$ can be expanded in a Fourier series as,

$$x_p[n] = \sum_{k=0}^{N-1} a_k e^{j\frac{2\pi kn}{N}} \dots \dots \dots (7.3)$$

where, $n = 0, 1, \dots, N-1$, with Fourier series coefficients,

$$a_k = \frac{1}{N} \sum_{n=0}^{N-1} x_p[n] e^{-\frac{j2\pi kn}{N}} \dots \dots \dots (7.4)$$

where, $k = 0, 1, \dots, N-1$

Upon comparing equation 7.2 and equation 7.4, we get

$$a_k = \frac{1}{N} X(e^{j\frac{2\pi k}{N}}) \quad , \quad k = 0, 1, \dots, N-1$$

then from equation 7.3,

$$x_p[n] = \frac{1}{N} \sum_{k=0}^{N-1} X(e^{j\frac{2\pi k}{N}}) e^{j\frac{2\pi kn}{N}} \quad \text{where } n = 0, 1, 2, \dots, N-1$$

This relationship provides the reconstruction of the periodic signal $x_p[n]$ from the samples of the spectrum $X(e^{j\frac{2\pi k}{N}})$. But it does not mean that we can recover $X(e^{j\omega})$ or $x[n]$ from the samples. In fact, to achieve this we need to establish a relationship between $x_p[n]$ and $x[n]$.

Now since $x_p[n]$ is the periodic extension of $x[n]$ as given by following expression:

$$x_p[n] = \sum_{l=-\infty}^{\infty} x[n - lN]$$

Then, it is obvious that $x[n]$ can be recovered from $x_p[n]$ if there is no aliasing in the time-domain. This means that if $x[n]$ is time-limited to less than the period N of $x_p[n]$. From figure 7.2 shown below, we observe that when $N \geq L$ (length of aperiodic sequence $x[n]$), then,

$$x[n] = x_p[n] \quad 0 \leq n \leq N-1$$

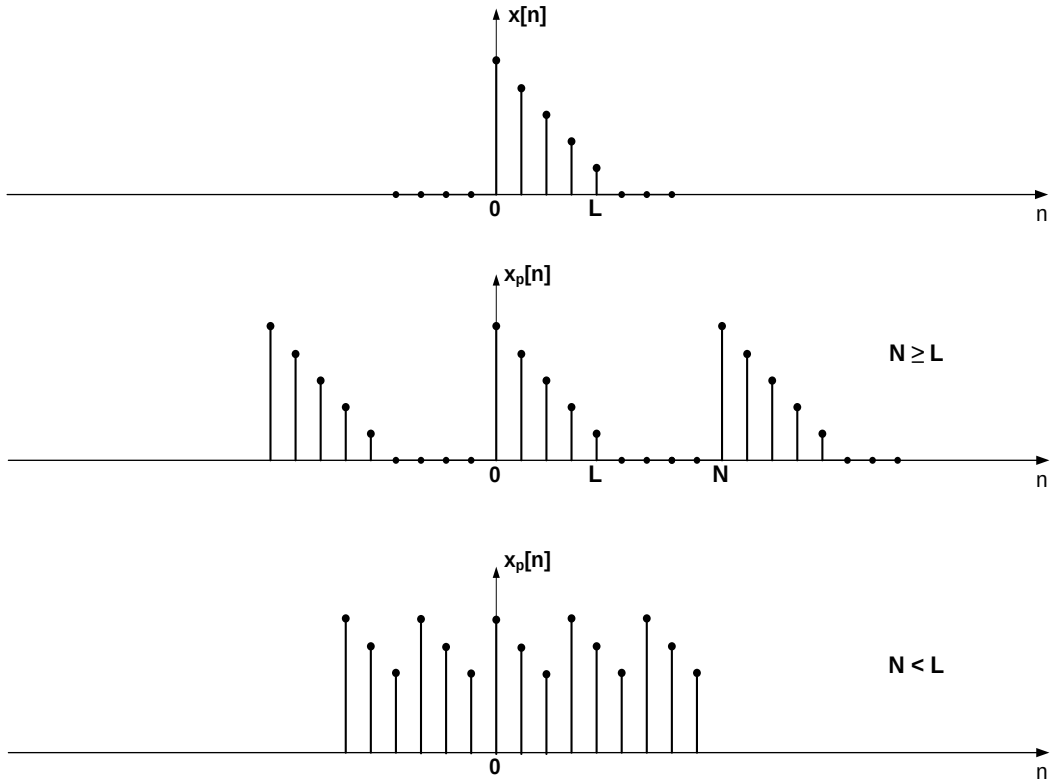


Figure 7.2: Aperiodic sequence $x[n]$ of length L and its periodic extension for $N \geq L$ (No aliasing) and $N < L$ (aliasing)

So, $x[n]$ can be recovered from $x_p[n]$. On the other hand, if $N < L$, it is not possible to recover $x[n]$ from $x_p[n]$ due to time domain aliasing as shown in figure 7.2. So in conclusion, one should choose N favoring the relation $N \geq L$ to prevent time domain aliasing in DFT calculation.

Discrete Fourier Transform (DFT) Pairs

$$X(k) = \sum_{n=0}^{N-1} x[n] e^{-\frac{j2\pi kn}{N}} \text{ for } k = 0, 1, \dots, N-1 \dots \dots \dots \quad \text{DFT Equation}$$

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{\frac{j2\pi kn}{N}} \text{ for } n = 0, 1, \dots, N-1 \dots \dots \dots \quad \text{IDFT Equation}$$

IDFT implies Inverse Discrete Fourier Transform.

DFT as a linear transformation

The DFT of discrete sequence $x[n]$ is denoted by $X(k)$, it is given by,

$$X(k) = \sum_{n=0}^{N-1} x[n] e^{-\frac{j2\pi kn}{N}}$$

Where, $k = 0, 1, 2, \dots, N-1$

Since, thus summation is taken for N points; it is called as “ N point DFT” and we can obtain discrete sequence $x[n]$ from its DFT. It is called as inverse discrete Fourier transforms (IDFT). It is given by,

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{\frac{j2\pi kn}{N}}$$

Where, $n = 0, 1, \dots, N-1$

Now, we will define the new term w_N as,

$$w_N = e^{-\frac{j2\pi}{N}}$$

This is called as **twiddle factor**. Twiddle factor makes the computation of DFT easier and faster.

Using twiddle factor, we can write equation of DFT and IDFT as under.

$$X(k) = \sum_{n=0}^{N-1} x[n]w_n^{kn} \quad \text{Where } k = 0, 1, 2, \dots, N-1$$

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X(k)w_n^{-kN} \quad \text{Where } n = 0, 1, 2, \dots, N-1$$

Let us define an N- point vector. x_N of frequency samples, and an $N \times N$ matrix W_N as,

Input signal

↓

 $x_N = \begin{bmatrix} x[0] \\ x[1] \\ \vdots \\ x[N-1] \end{bmatrix}$

DFT of the given signal

↙

 $X_N = \begin{bmatrix} X(0) \\ X(1) \\ \vdots \\ X(N-1) \end{bmatrix}$

Twiddle factor

↘

 $W_N = \begin{bmatrix} 1 & 1 & 1 & 1 & \dots & 1 \\ 1 & W_N^1 & W_N^2 & W_N^3 & \dots & W_N^{N-1} \\ 1 & W_N^2 & W_N^4 & W_N^6 & \dots & W_N^{2(N-1)} \\ \vdots & \vdots & \vdots & \vdots & \dots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \dots & \vdots \\ 1 & W_N^{N-1} & W_N^{2(N-1)} & W_N^{3(N-1)} & \dots & W_N^{(N-1)(N-1)} \end{bmatrix}$

With these definitions, the N – point DFT may be expressed in matrix form as.

$$X_N = W_N x_N$$

Where, W_N is the matrix of linear transformation.

Example 7.2: Compute the DFT of the 4 – point sequence $x[n] = [0 \ 1 \ 2 \ 3]$.

Solution:

We have 4-point DFT in matrix form as:

$$\begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & w_4^1 & w_4^2 & w_4^3 \\ 1 & w_4^2 & w_4^4 & w_4^6 \\ 1 & w_4^3 & w_4^6 & w_4^9 \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ x[3] \end{bmatrix}$$

We know,

$$w_N = e^{-\frac{j2\pi}{N}} \quad (\text{twiddle factor})$$

$$W_4 = e^{-\frac{j\pi}{2}}$$

$$W_4^1 = e^{-\frac{j\pi}{2}} = -j$$

So,

$$\begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & w_4^1 & w_4^2 & w_4^3 \\ 1 & w_4^2 & w_4^4 & w_4^6 \\ 1 & w_4^3 & w_4^6 & w_4^9 \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ x[3] \end{bmatrix}$$

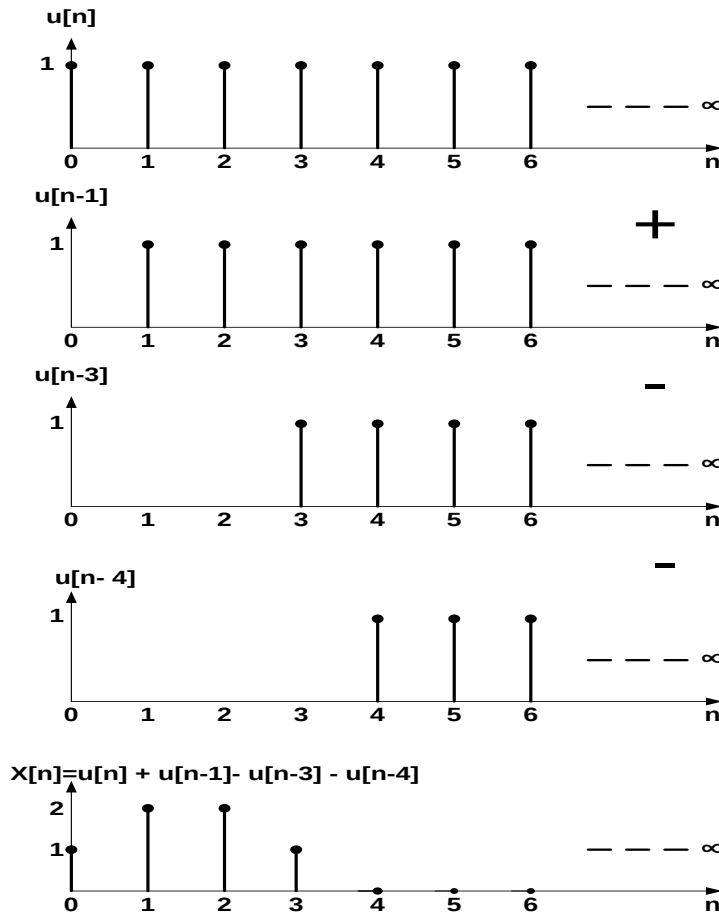
$$\begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 0-j-2+3j \\ 0-1+2-3 \\ 0+j-2-3j \end{bmatrix}$$

Therefore,

$$\begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \end{bmatrix} = \begin{bmatrix} 6 \\ -2 + 2j \\ -2 \\ -2 - 2j \end{bmatrix}$$

Example 7.9 Determine the 4-point DFT of the signal $x[n]=u[n]+ u[n-1]- u[n-3] - u[n-4]$.

Solution: First of all let us compute the elements of the discrete time sequence $x[n]$ graphically.



Hence $x[n] = [1, 2, 2, 1]$

We have,

$$\begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & w_4^1 & w_4^2 & w_4^3 \\ 1 & w_4^2 & w_4^4 & w_4^6 \\ 1 & w_4^3 & w_4^6 & w_4^9 \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ x[3] \end{bmatrix}$$

We know,

$$w_N = e^{-\frac{j2\pi}{N}} \quad (\text{twiddle factor})$$

$$W_4 = e^{-\frac{j\pi}{2}}$$

$$W_4^1 = e^{-\frac{j\pi}{2}} = -j$$

So,

$$\begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & w_4^1 & w_4^2 & w_4^3 \\ 1 & w_4^2 & w_4^4 & w_4^6 \\ 1 & w_4^3 & w_4^6 & w_4^9 \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ x[3] \end{bmatrix}$$

$$\begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 0-j-2+3j \\ 0-1+2-3 \\ 0+j-2-3j \end{bmatrix}$$

Therefore,

$$\begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \end{bmatrix} = \begin{bmatrix} 6 \\ -1-j \\ 0 \\ -1+j \end{bmatrix}$$

Properties of the DFT

We introduced the DFT as a set of N samples $X(k)$ of the Fourier transform $X(e^{j\omega})$ for a finite duration sequence $x[n]$ of length $L \leq N$. The sampling of $X(e^{j\omega})$ occurs at the N equally spaced frequencies $\omega_k = 2\pi k/N$, $k = 0, 1, \dots, N-1$, we also demonstrated that the N samples $X(k)$ uniquely represent the sequence $x[n]$ in the frequency domain. We have,

$$X(k) = \sum_{n=0}^{N-1} x[n] w_N^{kn} \quad \text{Where } k = 0, 1, 2, \dots, N-1 \dots \dots \dots \mathbf{DFT}$$

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] w_N^{-kN} \quad \text{Where } n = 0, 1, 2, \dots, N-1 \dots \dots \dots \mathbf{IDFT}$$

$$\text{Where, } w_N = e^{-\frac{j2\pi}{N}}$$

a)Periodicity :

If $x[n]$ and $X(k)$ are an N -point DFT pairs, then

$$X(k + N) = X(k) \quad \text{for all } k$$

$$X[n+N] = x[n] \quad \text{for all } n.$$

Proof: We have DFT of $x[n]$ as,

$$X(k) = \sum_{n=0}^{N-1} x[n] e^{-\frac{j2\pi kn}{N}} \quad \text{Where } k = 0, 1, 2, \dots, N-1$$

So,

$$\begin{aligned} X(k + N) &= \sum_{n=0}^{N-1} x[n] e^{-\frac{j2\pi(k+N)n}{N}} \\ &= \sum_{n=0}^{N-1} x[n] e^{-\frac{j2\pi kn}{N}} e^{-\frac{j2\pi Nn}{N}} \\ &= \sum_{n=0}^{N-1} x[n] e^{-\frac{j2\pi kn}{N}} e^{-j2\pi n} \\ &= \sum_{n=0}^{N-1} x[n] e^{-\frac{j2\pi kn}{N}} \quad [\because e^{-j2\pi n} = 1] \end{aligned}$$

$$\therefore X(k + N) = X(k)$$

Similarly, We have IDFT as,

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{\frac{j2\pi kn}{N}}$$

So,

$$x[n + N] = \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{\frac{j2\pi k(n+N)}{N}}$$

$$\begin{aligned}
&= \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{\frac{j2\pi kn}{N}} e^{\frac{j2\pi kN}{N}} \\
&= \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{\frac{j2\pi kn}{N}} \cdot e^{j2\pi k} \\
&= \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{\frac{j2\pi kn}{N}}
\end{aligned}$$

$$\therefore x[n + N] = x[n]$$

Example 7.3 If the DFT of any sequence $X(k)$ is given as $[1, -2+7j, 3, -2-7j]$, find $X(15)$.

Solution: Given,

$$X(0) = 1, X(1) = -2+7j, X(2) = 3, \text{ and } X(3) = -2-7j$$

Now from the periodicity property we know that,

$$X(k + N) = X(k) \quad \text{for all } k.$$

$$\text{Therefore, } X[3] = X[3+4]$$

$$\text{Therefore, } X[7] = X[3] = -2-7j$$

$$\text{Again, } X[7+4] = X[7]$$

$$\text{Therefore, } X[7] = X[11] = -2-7j$$

$$\text{Again, } X[11+4] = X[11]$$

$$\text{Therefore, } X[15] = X[11] = -2-7j$$

b)Linearity :

$$\text{If } x_1[n] \xleftrightarrow[N]{\text{DFT}} X_1(k) \quad \text{and}$$

$$x_2[n] \xleftrightarrow[N]{\text{DFT}} X_2(k)$$

then,

$$a x_1[n] + b x_2[n] \xleftrightarrow[N]{\text{DFT}} aX_1(k) + bX_2(k)$$

Proof:

Applying DFT as:

$$\begin{aligned}
 \text{DFT} \{ a x_1[n] + b x_2[n] \} &= \sum_{n=0}^{N-1} \{ a x_1[n] + b x_2[n] \} e^{-\frac{j2\pi kn}{N}} \\
 &= a \sum_{n=0}^{N-1} x_1[n] e^{-\frac{j2\pi kn}{N}} + b \sum_{n=0}^{N-1} x_2[n] e^{-\frac{j2\pi kn}{N}} \\
 &= aX_1(k) + bX_2(k)
 \end{aligned}$$

3) Multiplication of two DFT and circular convolution:

Statement:

The discrete Fourier transform of circular convolution of two discrete time sequences is equal to the product of discrete Fourier transform of individual sequences

$$\begin{aligned}
 \text{If } x_1[n] &\xleftrightarrow[N]{\text{DFT}} X_1(k) \quad \text{and} \\
 x_2[n] &\xleftrightarrow[N]{\text{DFT}} X_2(k)
 \end{aligned}$$

then,

$$x_1[n] \circledast x_2[n] \xleftrightarrow[N]{\text{DFT}} X_1(k) X_2(k)$$

Proof:

Suppose that we have two finite – duration sequences of length N, $x_1[n]$ and $x_2[n]$

Their respective N – point DFT are

$$X_1(k) = \sum_{n=0}^{N-1} x_1[n] e^{-\frac{j2\pi kn}{N}} \quad \text{for, } k = 0, 1, \dots, N-1$$

$$X_2(k) = \sum_{n=0}^{N-1} x_2[n] e^{-\frac{j2\pi kn}{N}} \quad \text{for, } k = 0, 1, \dots, N-1$$

If we multiply the two DFT's together, the result is also a DFT, i.e. say $X_3(k)$, of the sequence $x_3[n]$ of length N .

Let us determine the relationship between $x_3[n]$ and the sequence $x_1[n]$ and $x_2[n]$

We have,

$$X_3(k) = X_1(k) \cdot X_2(k)$$

The IDFT of $X_3[k]$ is,

$$\begin{aligned} x_3[m] &= \frac{1}{N} \sum_{k=0}^{N-1} X_3(k) e^{\frac{j2\pi km}{N}} \text{ for, } n = 0, 1 \dots \dots, N-1 \\ &= \frac{1}{N} \sum_{k=0}^{N-1} X_1(k) X_2(k) e^{\frac{j2\pi km}{N}} \end{aligned}$$

Suppose that we substitute for $X_1[k]$ and $X_2[k]$ using the DFT's we obtain,

$$\begin{aligned} x_3[m] &= \frac{1}{N} \sum_{k=0}^{N-1} \left[\sum_{n=0}^{N-1} x_1[n] e^{-\frac{j2\pi kn}{N}} \right] \left[\sum_{l=0}^{N-1} x_2[l] e^{-\frac{j2\pi kl}{N}} \right] e^{\frac{j2\pi km}{N}} \\ &= \frac{1}{N} \sum_{n=0}^{N-1} [x_1[n]] \left[\sum_{l=0}^{N-1} x_2[l] \right] \sum_{k=0}^{N-1} e^{\frac{j2\pi k(m-n-l)}{N}} \dots \dots \dots (7.5) \end{aligned}$$

The inner sum in the brackets has the form

$$\sum_{k=0}^{N-1} a_k = \begin{cases} N & \text{for } a = 1 \\ \frac{1-a^N}{1-a} & \text{for } a \neq 1 \end{cases} \dots \dots \dots (7.6)$$

Where a is defined as,

$$a = e^{\frac{j2\pi(m-n-l)}{N}}$$

We observe that $a = 1$ when $m - n - l$ is a multiple of N . on the other hand, $a^N = 1$ for any value of $a \neq 0$ and consequently equation (7.6) reduces to,

$$\sum_{k=0}^{N-1} a_k = \begin{cases} N & \text{for } l = m - n + PN = ((m - n))_N, P \text{ is an integer} \\ 0 & \text{Otherwise} \end{cases}$$

If we substitute the result in equation (7.5)

$$x_3[m] = \sum_{n=0}^{N-1} x_1[n] x_2[(m-n)]_N \quad m = 0, 1, \dots, N-1$$

$$= x_1[n] \circledast_N x_2[n]$$

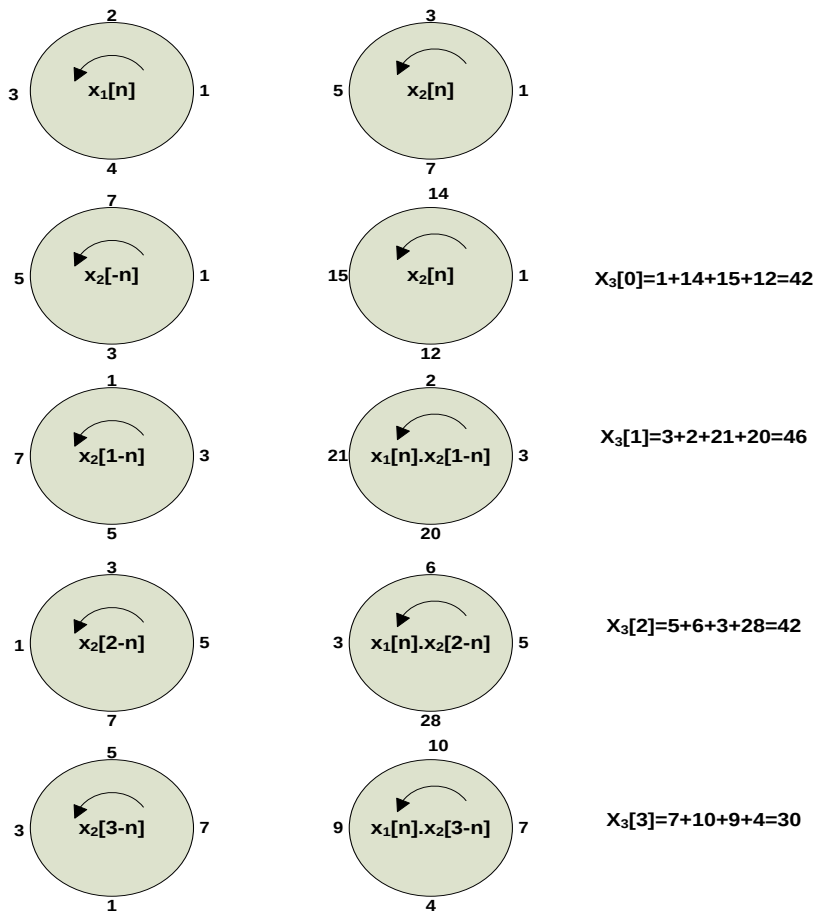
So,

$$x_1[n] \circledast_N x_2[n] \xleftrightarrow[N]{\text{DFT}} X_1(k) X_2(k)$$

Thus we conclude that multiplication of the DFTs of two sequences is equivalent to the circular convolution of the sequences in time domain.

Example 7.6: Perform the circular convolution of the following two sequences $x_1[n]$ and $x_2[n]$ where, $x_1[n] = \{1, 2, 3, 4\}$ and $x_2[n] = \{1, 3, 5, 7\}$.

Solution:



Hence, $x_3[m] = [42 \ 72 \ 49 \ 30]$

Zero padding:

The process of adding zeroes to the original sequence for the efficient computation of DFT is called as zero padding.

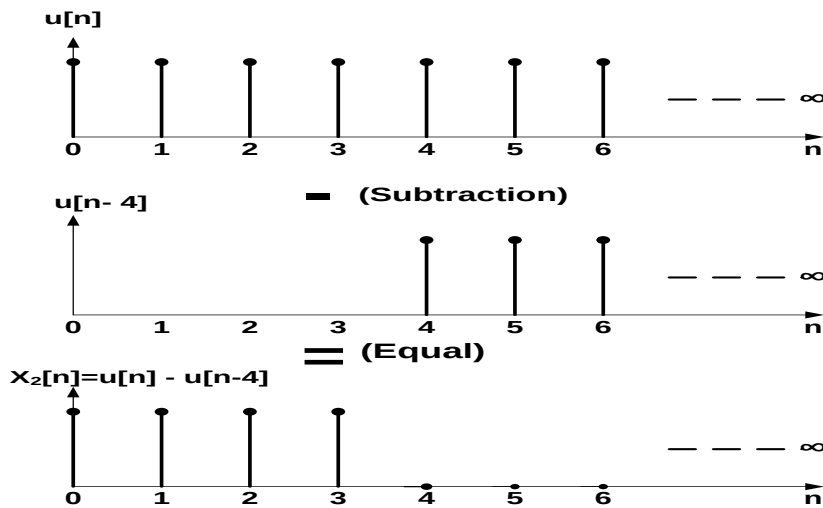
Note that zero padding does not provide any additional information about the spectrum $X(j\omega)$ of the sequence $x[n]$. However padding the sequence $x[n]$ and computing an N – point DFT results in a “better display” of the Fourier transform $X(j\omega)$.

Example 7.7 Why do you need DFT although you have DTFT? Find circular convolution between $x_1[n] = \{1, 2\}$ and $x_2[n] = u[n] - u[n - 4]$.

Solution:

To perform spectral analysis on a discrete time signal $x[n]$ we convert the time domain sequence to an equivalent frequency – domain representation. We know that such a representation is given by the Discrete Time Fourier transform $X(e^{j\omega})$ of the sequence $x[n]$. However $X(e^{j\omega})$ is a continuous function of frequency and therefore, it is not a computationally convenient representation of the sequence $x[n]$. Now to overcome these discrepancies we consider the representation of a sequence $x[n]$ by sample of its spectrum $X(e^{j\omega})$. Such a frequency – domain representation leads to discrete Fourier transform (DFT), which is a powerful computational tool for perform frequency analysis of discrete – time signals.

To find out the circular convolution between $x_1[n]$ and $x_2[n]$ first of all we have to find out the elements of $x_2[n]$. The steps of finding $x_2[n]$ are shown below:



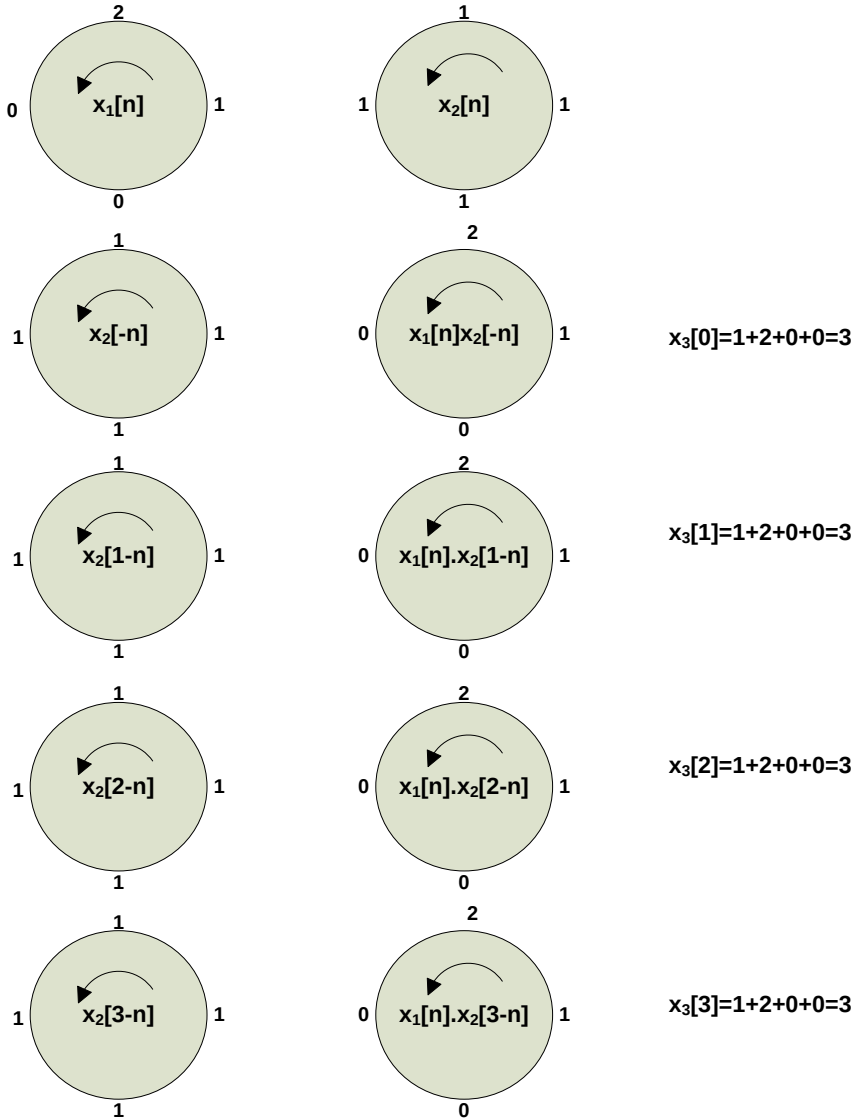
So, $x_2[n] = \{1, 1, 1, 1\}$

Now in order to find the circular convolution we need to pad zeroes to the sequence $x_1[n]$ so that length of $x_1[n]$ and $x_2[n]$ becomes equal.

Therefore $\mathbf{x}_1[\mathbf{n}] = \{ 1, 2, 0, 0 \}$

Let the circular convolution of $x_1(n)$ & $x_2(n)$ is $x_3[m]$

$$x_3[m] = \sum_{n=0}^{N-1} x_1[n]x_2[m-n]_4 \quad m = 0, 1, \dots, N-1$$



Hence, $\mathbf{x}_3[\mathbf{m}] = \{3, 3, 3, 3\}$

Difference between Linear and Circular convolution:

Linear Convolution	Circular Convolution
1. If x_1 is sequence of length l_1 , x_2 is sequence of length l_2 , and then length of output of linear convolution is l_1+l_2-1 .	1. If x_1 is sequence of length l_1 , x_2 is sequence of length l_2 , and then length of output of circular convolution is $l_1=l_2$.
2. It can be operate on sequence of different length.	2. It must be operate on equal length sequence.
3. The output of convolution sum includes linear flip and shift given as $y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$	3. The output of circular convolution includes circular version as $y[n] = \sum_{k=-N}^N x[k]h[(m-n)]_N$
4. $x_1[n] \otimes x_2[n] \xleftrightarrow{\text{DTFT}} X_1(e^{j\omega}) \cdot X_2(e^{j\omega})$	4. $x_1[n] \otimes x_2[n] \xleftrightarrow{\text{DFT}} X_1[k] \cdot X_2[k]$

d) Time Reversal:

$$\text{If } x[n] \xleftrightarrow[N]{DFT} X(k)$$

then,

$$x[(-n)]_N = x[N-n] \xleftrightarrow[N]{DFT} X((-k))_N = X(N-k)$$

Proof:

$$\text{DFT} \{ x[(-n)]_N \} = \text{DFT} \{ x[N-n] \} = \sum_{n=0}^{N-1} x[N-n] e^{-\frac{j2\pi kn}{N}}$$

Put $m=N-n$, then

$$\begin{aligned} \text{DFT} \{ m \} &= \text{DFT} \{ x[N-n] \} = \sum_{m=1}^N x[m] e^{-\frac{j2\pi k(N-m)}{N}} = \sum_{m=0}^{N-1} x[m] e^{-\frac{j2\pi k(N-m)}{N}} \\ &= \sum_{m=0}^{N-1} x[m] e^{-j2\pi k} \cdot e^{\frac{j2\pi km}{N}} = \sum_{m=0}^{N-1} x[m] e^{\frac{j2\pi km}{N}} \end{aligned}$$

$$\begin{aligned}
&= \sum_{m=0}^{N-1} x[m] e^{-\frac{j2\pi m(N-k)}{N}} \\
&= X(N-k) = X((-k))_N
\end{aligned}$$

a) Circular time shift of a sequence:

$$\text{If } x[n] \xleftrightarrow[N]{\text{DFT}} X(k)$$

then,

$$x[(n-l)]_N \xleftrightarrow[N]{\text{DFT}} X(k) e^{-\frac{j2\pi kl}{N}}$$

Proof:

$$\begin{aligned}
\text{DFT} \{ x[n-l]_N \} &= \sum_{n=0}^{N-1} x[n-l]_N e^{-\frac{j2\pi kn}{N}} \\
&= \sum_{n=0}^{l-1} x[n-l]_N e^{-\frac{j2\pi kn}{N}} + \sum_{n=l}^{N-1} x[n-l]_N e^{-\frac{j2\pi kn}{N}} \dots \dots (7.7)
\end{aligned}$$

We know that, $x[n-l]_N = x[N-l+n]$ and if $m=N-l+n$, then,

$$\begin{aligned}
\sum_{n=0}^{l-1} x[n-l]_N e^{-\frac{j2\pi kn}{N}} &= \sum_{n=0}^{l-1} x[N-l+n]_N e^{-\frac{j2\pi kn}{N}} = \sum_{m=N-l}^{N-1} x[m]_N e^{-\frac{j2\pi k(m+l-N)}{N}} \\
\therefore \sum_{n=0}^{l-1} x[n-l]_N e^{-\frac{j2\pi kn}{N}} &= \sum_{m=N-l}^{N-1} x[m]_N e^{-\frac{j2\pi k(m+l)}{N}} \dots \dots \dots (7.8)
\end{aligned}$$

Again,

$$\begin{aligned}
\sum_{n=l}^{N-1} x[n-l]_N e^{-\frac{j2\pi kn}{N}} &= \sum_{n=l}^{N-1} x[N-l+n]_N e^{-\frac{j2\pi kn}{N}} = \sum_{m=N}^{2N-l-1} x[m]_N e^{-\frac{j2\pi k(m+l-N)}{N}} \\
\therefore \sum_{n=l}^{N-1} x[n-l]_N e^{-\frac{j2\pi kn}{N}} &= \sum_{m=N}^{2N-l-1} x[m]_N e^{-\frac{j2\pi k(m+l)}{N}} \\
&= \sum_{m=0}^{N-l-1} x[m]_N e^{-\frac{j2\pi k(m+l)}{N}} \dots \dots \dots (7.9)
\end{aligned}$$

Now substituting from equations (7.8) to (7.9) in equation (7.7), we get,

$$\begin{aligned}
 \text{DFT } \{x[n-l]_N\} &= \sum_{m=N-l}^{N-1} x[m]_N e^{-\frac{j2\pi k(m+l)}{N}} + \sum_{m=0}^{N-l-1} x[m]_N e^{-\frac{j2\pi k(m+l)}{N}} \\
 &= \sum_{m=0}^{N-1} x[m]_N e^{-\frac{j2\pi k(m+l)}{N}} = \left[\sum_{m=0}^{N-1} x[m]_N e^{-\frac{j2\pi km}{N}} \right] e^{-\frac{j2\pi kl}{N}} \\
 &= X(k) e^{-\frac{j2\pi kl}{N}}
 \end{aligned}$$

b) Circular frequency shift:

$$\text{If } x[n] \xleftrightarrow[N]{DFT} X(k)$$

then,

$$x[n]e^{\frac{j2\pi ln}{N}} \xleftrightarrow[N]{DFT} X((k-l))_N$$

Proof:

$$\begin{aligned}
 \text{DFT } \left\{ x[n]e^{\frac{j2\pi ln}{N}} \right\} &= \text{IDFT of } X((k-l))_N \\
 &= \frac{1}{N} \sum_{k=0}^{N-1} X((k-l))_N e^{\frac{j2\pi kn}{N}} \\
 &= \frac{1}{N} \sum_{k=0}^{l-1} X(N+k-l) e^{\frac{j2\pi kn}{N}} \\
 &\quad + \frac{1}{N} \sum_{k=l}^{N-1} X(N+k-l) e^{\frac{j2\pi kn}{N}} \\
 &= I_1 + I_2
 \end{aligned}$$

Where,

$$I_1 = \frac{1}{N} \sum_{k=0}^{l-1} X(N+k-l) e^{\frac{j2\pi kn}{N}}$$

Put $m=N+k-l$, then

$$I_1 = \frac{1}{N} \sum_{m=N-1}^{N-1} X(m) e^{\frac{j2\pi(m+1-N)n}{N}} = \frac{1}{N} \sum_{m=N-1}^{N-1} X(m) e^{\frac{j2\pi(m+1)n}{N}}$$

Again,

$$I_2 = \frac{1}{N} \sum_{k=1}^{N-1} X(N+k-l) e^{\frac{j2\pi kn}{N}}$$

Put $m=N+k-l$, then

$$I_2 = \frac{1}{N} \sum_{m=N}^{2N-l-1} X(m) e^{\frac{j2\pi(m+1-N)n}{N}} = \frac{1}{N} \sum_{m=0}^{N-l-1} X(m) e^{\frac{j2\pi(m+1)n}{N}}$$

So,

$$\begin{aligned} \text{DFT} \left\{ x[n] e^{\frac{j2\pi ln}{N}} \right\} &= I_1 + I_2 \\ &= \frac{1}{N} \sum_{m=N-1}^{N-1} X(m) e^{\frac{j2\pi(m+1)n}{N}} + \frac{1}{N} \sum_{m=0}^{N-l-1} X(m) e^{\frac{j2\pi(m+1)n}{N}} \\ &= \frac{1}{N} \sum_{m=0}^{N-1} X(m) e^{\frac{j2\pi(m+1)n}{N}} = \left[\frac{1}{N} \sum_{m=0}^{N-1} X(m) e^{\frac{j2\pi mn}{N}} \right] e^{\frac{j2\pi ln}{N}} \\ &= x[n] e^{\frac{j2\pi ln}{N}} \end{aligned}$$

c) Complex conjugate property:

$$\text{If } x[n] \xleftrightarrow[N]{DFT} X[k]$$

then,

$$x^*[n] \xleftrightarrow[N]{DFT} X^*[-k]_N = X^*(N-k)$$

Proof:

$$\begin{aligned} \text{DFT} \{ x^*[n] \} &= \sum_{n=0}^{N-1} x^*[n] e^{-\frac{j2\pi kn}{N}} = \sum_{n=0}^{N-1} x^*[n] e^{\frac{j2\pi(N-k)n}{N}} \\ &= \left[\sum_{n=0}^{N-1} x[n] e^{-\frac{j2\pi(N-k)n}{N}} \right]^* \end{aligned}$$

$$=X^*(N-k) = X^*[(-k)]_N$$

d) Multiplication of two sequence:

$$\text{If } x_1[n] \xleftrightarrow[N]{DFT} X_1[k] \quad \text{and}$$

$$x_2[n] \xleftrightarrow[N]{DFT} X_2[k]$$

then,

$$x_1[n] \cdot x_2[n] \xleftrightarrow[N]{DFT} 1/N \{X_1(k) \odot X_2(k)\}$$

Proof:

$$\begin{aligned} \text{DFT} \{x_1[n]x_2[n]\} &= \sum_{n=0}^{N-1} x_1[n]x_2[n]e^{-j2\pi kn/N} \\ &= \sum_{n=0}^{N-1} \left\{ \frac{1}{N} \sum_{l=0}^{N-1} X_1[l]e^{j2\pi ln/N} \right\} \left\{ \frac{1}{N} \sum_{p=0}^{N-1} X_2[p]e^{j2\pi pn/N} \right\} e^{-j2\pi kn/N} \\ &= \frac{1}{N^2} \sum_{l=0}^{N-1} X_1[l] \sum_{p=0}^{N-1} X_2[p] \sum_{n=0}^{N-1} e^{\frac{j2\pi ln(-k+l+p)}{N}} \dots \dots \dots (7.10) \end{aligned}$$

Where,

$$\sum_{n=0}^{N-1} e^{\frac{j2\pi ln(-k+l+p)}{N}} = \begin{cases} N & \text{if } -k+l+p \text{ is multiple of } N \\ 0 & \text{otherwise} \end{cases}$$

Here, $-k+l+p$ is multiple of N means, $-k+l+p = \lambda N \Rightarrow p = k-l + \lambda N = ((k-l))_N$

So, from equation 7.10,

$$\begin{aligned} \text{DFT} \{x_1[n]x_2[n]\} &= \frac{1}{N^2} \sum_{l=0}^{N-1} X_1[l] \sum_{p=0}^{N-1} X_2((k-l))_N \cdot N \\ &= \frac{1}{N} \sum_{l=0}^{N-1} X_1[l] \cdot X_2((k-l))_N \\ &= 1/N \{X_1(k) \odot X_2(k)\} \end{aligned}$$

Fast Fourier Transform (FFT)

As we have observed, the DFT plays an important role in many applications of digital signal processing. A major reason for its importance is the existence of efficient algorithms for computing the DFT. The Fast Fourier Transform (FFT) does not represent a transform different from DFT rather they are special algorithms for faster implementation of DFT. FFT requires a comparatively smaller number of arithmetic operations such as multiplications and additions than DFT that is the major reason why FFT requires lesser computational time than DFT. We use two algorithms namely, Decimation In Time FFT (DIT FFT) and Decimation In Frequency FFT (DIF FFT) for calculation of DFT.

i. Decimation In Time FFT (DITFFT):

a. 4-point DITFFT structure:

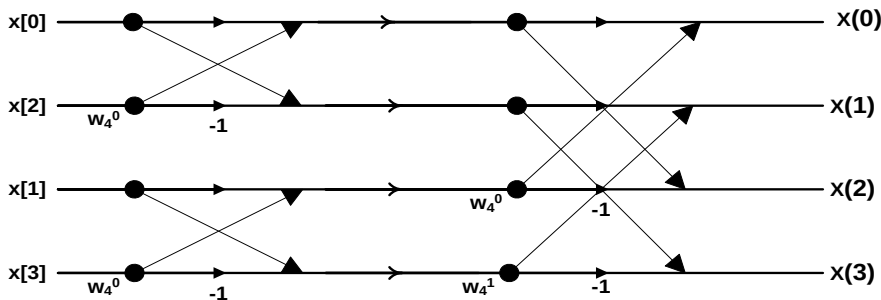


Figure 7.12: 4-point decimation in time (DIT) radix-2-FFT butterfly structure

b. 8-point DITFFT structure:

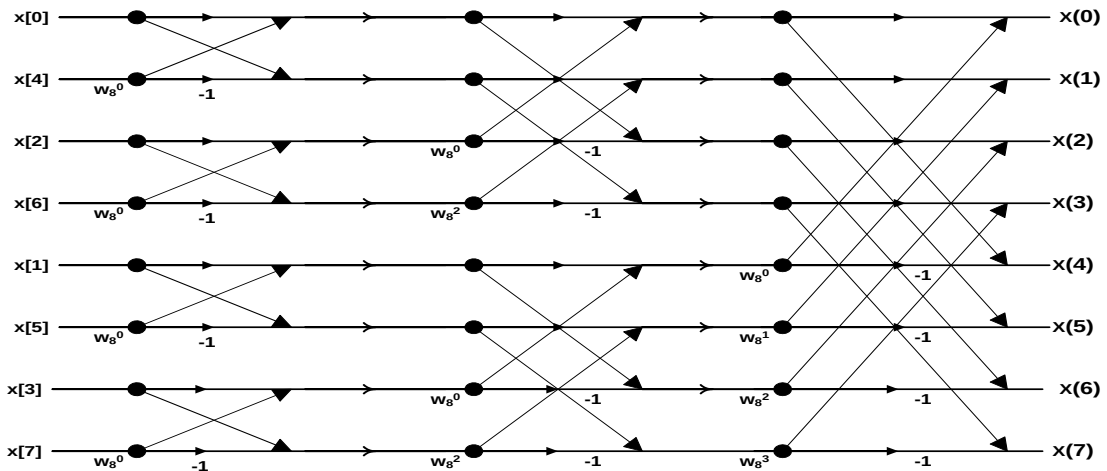


Figure 7.11: 8-point decimation in time (DIT) radix-2-FFT butterfly structure

Example 7.10 Compute the DFT of the sequence $x[n]=[0,1,2,3]$ using decimation in time FFT algorithm.

Solution:

Since it is a 4 point sequence we incorporate the figure 7.7 to compute the FFT

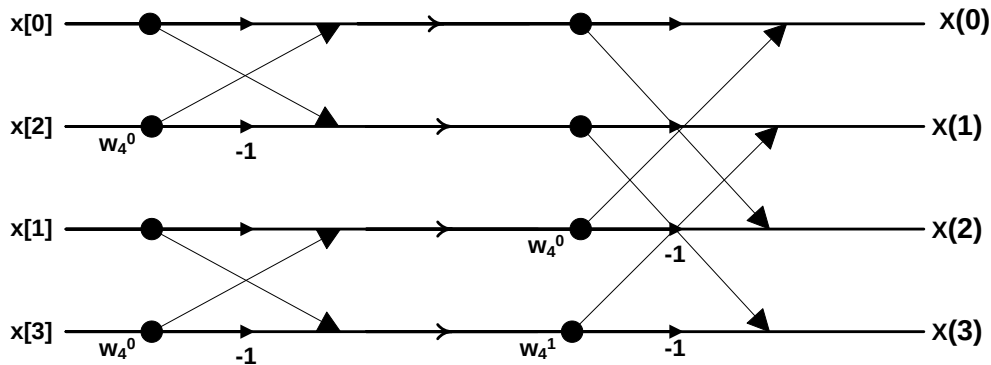


Figure 7.13: 4-point decimation in time (DIT) radix-2-FFT butterfly structure

Where $x[]$ represents the input sequence and $X()$ represents the DFT of the given sequence.

We have twiddle factor as,

$$w_N = e^{-\frac{j2\pi}{N}}$$

Then,

$$w_4 = e^{-\frac{j2\pi}{4}} = -j$$

So, $w_4^0 = 1$ and $w_4^1 = -j$

Let us begin finding the DFT starting from $X(0)$. Let's move all the input sequence towards $X(0)$.

$$X(0) = x[0] + w_4^0 x[2] + w_4^0 x[1] + w_4^0 x[3] = x[0] + x[2] + x[1] + x[3] = 0 + 1 + 2 + 3 = 6$$

Similarly for $X(1)$ to $X(3)$,

$$X(1) = x[0] - w_4^0 x[2] + w_4^1 x[1] - w_4^0 w_4^1 x[3] = x[0] - x[2] + (-j)x[1] - (-j)x[3]$$

$$= 0 - 2 - j + 3j = -2 + 2j$$

$$X(2) = x[0] + w_4^0 x[2] - w_4^0 x[1] - w_4^0 w_4^0 x[3] = 0 + 2 - 1 - 3 = -2$$

$$X(3) = x[0] - w_4^0 x[2] - w_4^1 x[1] + w_4^0 w_4^1 x[3] = 0 - 2 - (-j) + (-j)3 = -2 - 2j$$

Hence DFT of the given sequence is $X(k) = [6, -2+2j, -2, -2-2j]$

Example 7.12 Find the DFT of the signal $x[n] = [1,2,3,4,5]$ using DITFFT algorithm

Solution:

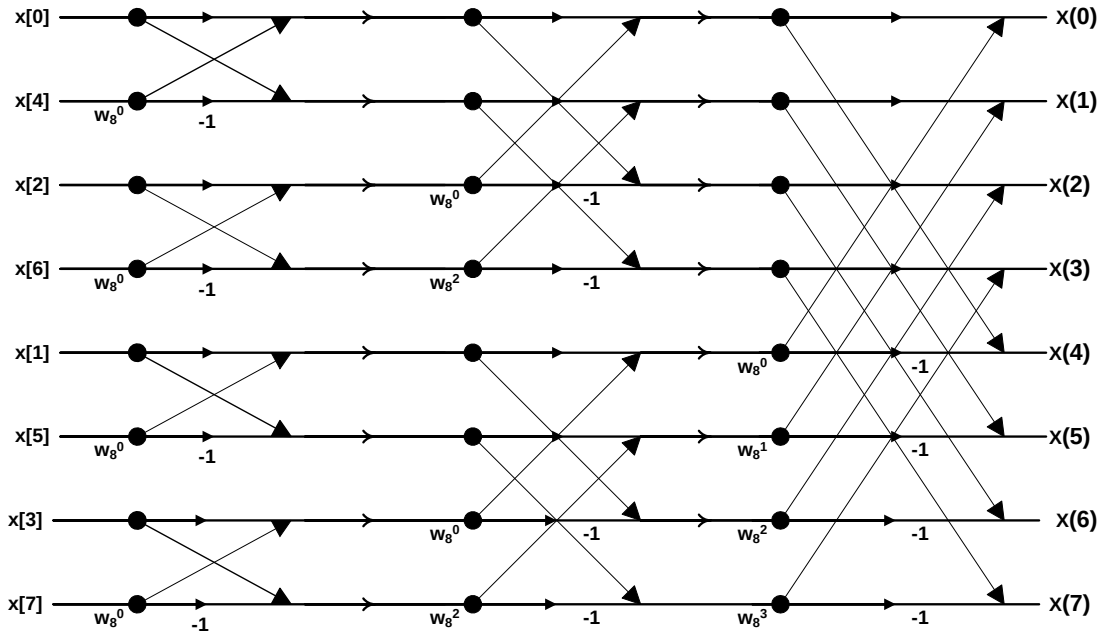


Figure 7.15: 8-point decimation in time (DIT) radix-2-FFT butterfly structure

$$\begin{aligned} X(1) &= x[0] - w_8^0 x[4] + w_8^2 x[2] - w_8^0 w_8^2 x[6] + w_8^1 x[1] - w_8^0 w_8^1 x[5] + w_8^2 w_8^1 x[3] - w_8^0 w_8^2 w_8^1 x[7] \\ &= 1 - 5 + 3(-j) - 0 + 2 \cdot \frac{1}{\sqrt{2}}(1-j) - 0 + 4(-j) \frac{1}{\sqrt{2}}(1-j) - 0 = -5.414 - 7.24j \end{aligned}$$

For simplicity don't care the segment associated part with $x[5]$, $x[6]$ and $x[7]$ because there value is zero.

$$\begin{aligned} X(2) &= x[0] + w_8^0 x[4] - w_8^0 x[2] + w_8^2 x[1] - w_8^0 w_8^2 x[3] \\ &= 1 + 5 - 3 - 2j + 4j = 3 + 2j \end{aligned}$$

$$\begin{aligned} X(3) &= x[0] - w_8^0 x[4] - w_8^2 x[2] + w_8^3 x[1] - w_8^2 w_8^3 x[3] \\ &= 1 - 5 + 3j + 2 \frac{1}{\sqrt{2}}(-1-j) - 4(-j) \frac{1}{\sqrt{2}}(-1-j) = -2.58 - 1.24j \end{aligned}$$

$$\begin{aligned} X(4) &= x[0] + w_8^0 x[4] + w_8^0 x[2] - w_8^0 x[1] - w_8^0 w_8^0 x[3] \\ &= 1 + 5 + 3 - 2 - 4 = 3 \end{aligned}$$

$$X(5) = x[0] - w_8^0 x[4] + w_8^2 x[2] - w_8^1 x[1] - w_8^2 w_8^1 x[3]$$

$$= 1 - 5 - 3j - 2 \cdot \frac{1}{\sqrt{2}}(1-j) - 4(-j) \frac{1}{\sqrt{2}}(1-j) = -2.58 + 1.24j$$

$$X(6) = x[0] + w_8^0 x[4] - w_8^0 x[2] - w_8^2 x[1] + w_8^0 w_8^2 x[3]$$

$$= 1 + 5 - 3 + 2j - 4j = 3 - 2j$$

$$X(7) = x[0] - w_8^0 x[4] + w_8^2 x[2] - w_8^1 x[1] - w_8^2 w_8^1 x[3]$$

$$= -5.414 + 7.24j$$

$$\text{Hence } X(k) = [15, -5.414 - 7.24j, 3 + 2j, -2.58 - 1.24j, 3, -2.58 + 1.24j, 3 - 2j, -5.414 + 7.24j]$$

My dear students' one important conclusion has been drawn from example 7.12 from which you can save your time in computing FFT.

If you are required to compute 8 point DFT using DITFFT algorithm then first of all find out $X(0), X(1), X(2), X(3)$ and $X(4)$ then list out two numbers such that their sum results to 8.

$$1+7=8 \quad 2+6=8 \quad 3+5=8 \quad 4+4=8$$

You will already have been computed $X(1)$ so there exists complex conjugate value at $k=7$ i.e. $X(7)$ contains complex conjugate value of $X(1)$.

In Example $X(1) = -5.414 - 7.24j$ so there exists complex conjugate for $X(7)$ i.e.

$$X(7) = -5.414 + 7.24j$$

$$\begin{array}{l|l|l} \text{Similarly } X(2) = 3 + 2j & X(3) = -2.58 - 1.24j & X(4) = 3 \\ X(6) = 3 - 2j & X(5) = -2.58 + 1.24j & X(4) = 3 \end{array}$$

Similar procedure can be followed in computing 4-point DFT, 16-point DFT and so on.

ii. Decimation in Frequency (DIF) FFT Algorithm

a. 4-point DIF FFT algorithm

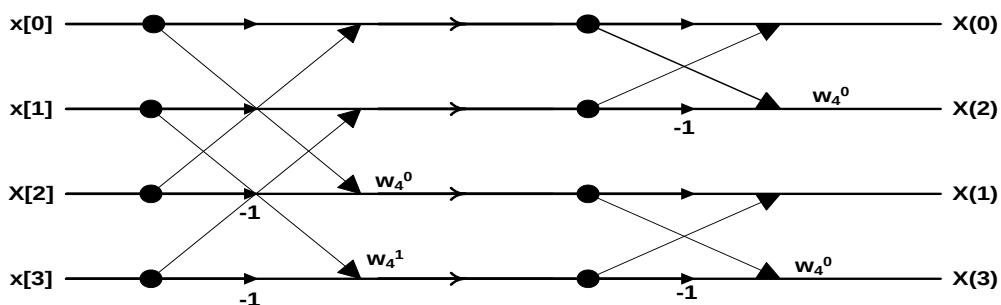


Figure 7.21: 4-point decimation in frequency (DIF) radix-2-FFT butterfly structure

a. 8-point DIF FFT algorithm

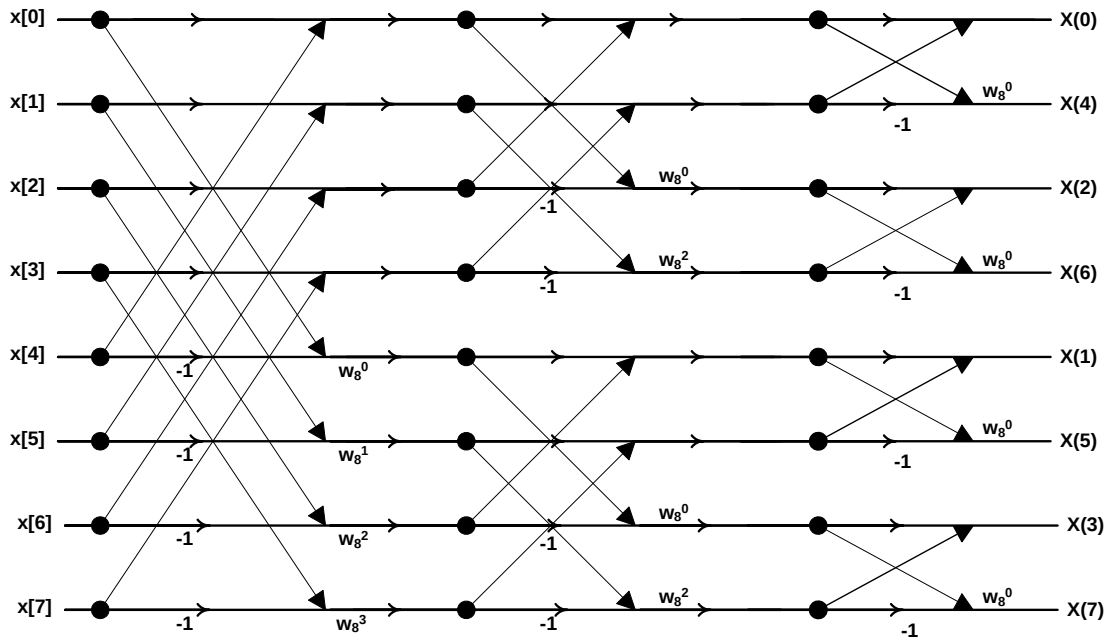


Figure 7.20: 8-point decimation in frequency (DIF) radix-2-FFT butterfly structure

Example 7.13 Find $X[k]$ for a given sequence $x[n]=[2,1,-2]$ using DIFFFT algorithm.

Solution:

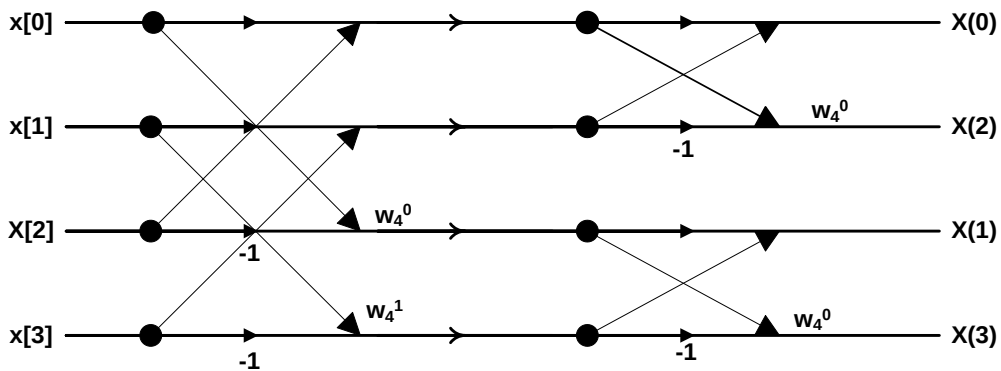


Figure 7.22: 4-point decimation in frequency (DIF) radix-2-FFT butterfly structure

$$X(0) = x[0] + x[1] + x[2] + x[3] = 1$$

$$\text{Similarly, } X(1) = w_4^0 x[0] + w_4^1 x[1] - w_4^0 x[2] - w_4^1 x[3] = 2 - j + 2 = 4 - j$$

$$X(2) = w_4^0 x[0] - w_4^0 x[1] + w_4^0 x[2] - w_4^0 x[3] = 2 - 1 - 2 = -1$$

We obtain complex conjugate value of $X(1)$ while we compute $X(3)$. Therefore $X(3) = 4 + j$

Example 7.14 Find the DFT of the signal $x[n] = [1,2,3,4,5]$ using DIFFFT algorithm

Solution:

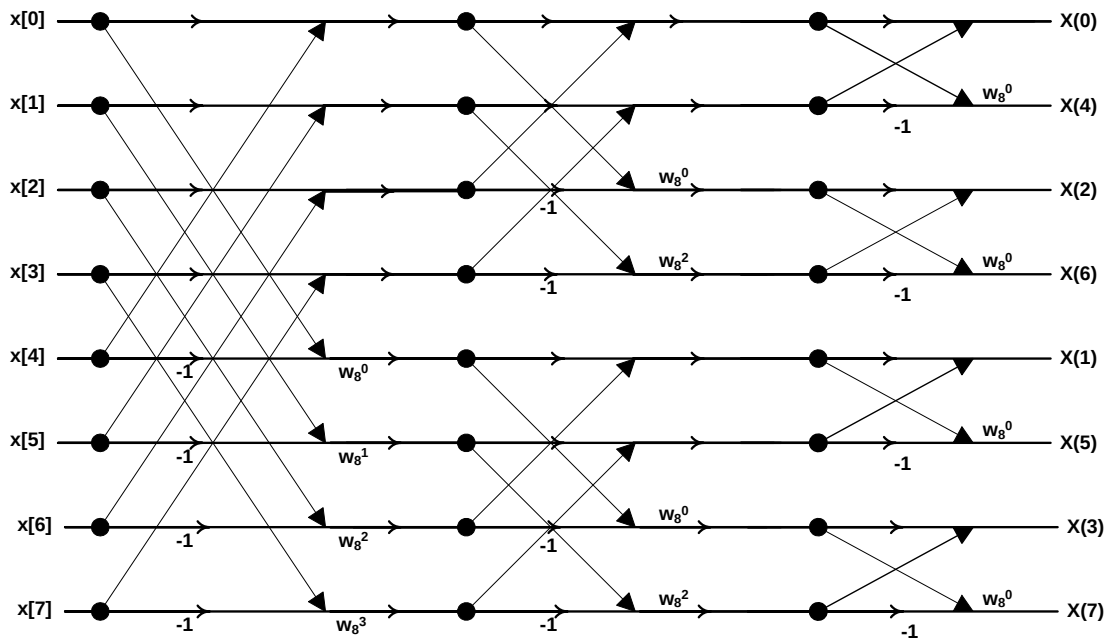


Figure 7.23: 8-point decimation in frequency (DIF) radix-2-FFT butterfly structure

Since the given sequence is not in 2's power we have to pad three zeroes in the input sequence which results in 8-point input sequence

Therefore $x[n] = [1, 2, 3, 4, 5, 0, 0, 0]$

Now to compute the 8-point DFT we accommodate the flow diagram 7.19

From figure 7.19 we find the DFT of the sequence

Suppose we have to find $X(0)$ then we move all the input sequence towards $X(0)$.

$$X(0)=x[0]+x[1]+x[2]+x[3]+x[4]+ x[5]+ x[6]+ x[7] =15$$

For simplicity don't care the segment associated part with $x[5]$, $x[6]$ and $x[7]$ because there value is zero.

Similarly,

$$X(4)= w_8^0 x[0] -w_8^0 x[1] + w_8^0 x[2] - w_8^0 x[3]+w_8^0 x[4] = 1-2+3+5-4 = 3$$

$$\begin{aligned} X(2)&= w_8^0 x[0] +w_8^2 x[1] -w_8^0 x[2] - w_8^2 x[3]+ w_8^0 x[4] \\ &= 1-2j-3 +4j +5 =3+2j \end{aligned}$$

$$X(1) = w_8^0 x[0] + w_8^1 x[1] + w_8^2 x[2] + w_8^3 x[3] - w_8^0 x[4] - w_8^1 x[5] - w_8^0 w_8^2 x[6] - w_8^3 x[7]$$

$$= 1 + 2 \cdot \frac{1}{\sqrt{2}}(1-j) + 3 \cdot (-j) + 4 \cdot (-j) \cdot \frac{1}{\sqrt{2}}(1-j) - 5 - 0 - 0 - 0 = -5.414 - 7.24j$$

$$X(5) = w_8^0 x[0] - w_8^1 x[1] + w_8^2 x[2] - w_8^3 x[3] - w_8^0 x[4]$$

$$= 1 - 5 - 3j - 2 \cdot \frac{1}{\sqrt{2}}(1-j) - 4(-j) \cdot \frac{1}{\sqrt{2}}(1-j) = -2.58 + 1.24j$$

Now we can simply write down the other point DFT and save your time in computing FFT.

If you are required to compute 8 point DFT using FFT algorithm then first of all find out $X(0), X(4), X(2), X(1)$ and $X(5)$ then list out two numbers such that their sum results to 8.

$$1+7=8 \quad 2+6=8 \quad 3+5=8 \quad 4+4=8$$

You will already have been computed $X(1)$ so there exists complex conjugate value at $k=7$ i.e. $X(7)$ contains complex conjugate value of $X(1)$.

In Example $X(1) = -5.414 - 7.24j$ so there exists complex conjugate for $X[7]$ i.e.

$$X(7) = -5.414 + 7.24j$$

$$\begin{array}{l} \text{Similarly } X(2) = 3+2j \quad \left| \quad X(3) = -2.58 - 1.24j \quad \left| \quad X(4) = 3 \right. \right. \\ X(6) = 3-2j \quad \left| \quad X(5) = -2.58 + 1.24j \quad \left| \quad X(4) = 3 \right. \right. \end{array}$$

Similar procedure can be followed in computing 4-point DFT, 16-point DFT and so on.

Speed improvement factor in FFT(Computational complexity of FFT):

The number of complex multiplication operations required by direct computation of DFT and IDFT is of order N^2 as there are N data points to calculate. In computer science, we may say they have algorithmic complexity of order N^2 and hence is not a very efficient method. If we can't do any better than this then the DFT will not be very useful for the majority of practical DSP (Digital Signal Processing) applications. However, there are a number of different FFT algorithms that enable calculating Fourier transform of the signal much faster than DFT.

As the name suggests, FFTs are algorithms for quick calculation of DFT. The FFT is a DFT algorithm which reduces the number of computations needed for N points from N^2 to $N \log_2 N$. The radix-2 algorithms are useful if N is a regular power of 2 ($N=2^v$). If we assume that algorithmic complexity provides a direct measure of execution time and that the relevant logarithm base is 2. Then as shown in table 7.2, ratio of execution time for DFT versus radix-2 FFT denoted as "Speed Improvement Factor" increases tremendously with increase in N .

Number of points N	Complex multiplication in direct computation of DFT(N ²)	Complex multiplication in FFT algorithm (N/2)log ₂ N	Speed improvement factor ($\frac{N^2}{N/2 \log_2 N}$)
4	16	4	4
8	64	12	5.3
16	256	32	8
32	1024	80	12.8
64	4096	192	21.3
128	16384	448	36.6
256	65356	1024	64
512	262144	2304	113.8
1024	1024576	5120	204.8

Table 7.2: Comparison of execution time, DFT and Radix-2 FFT

There are two different radix-2 algorithms, so called DITFFT and DIFFFT algorithm. Both of these rely on the recursive decomposition of

N point transforms into two $\frac{N}{2}$ transforms. This decomposition process can be applied to any composite (non prime) N. The method is particularly simple if N is divisible by 2 and if N is regular power of 2, the decomposition can be applied repeatedly until the trivial “1-point” transform is reached.

The decimation, however, causes shuffling in data. The entire process involves $P = \log_2 N$ stages of decimation, where each stage involves $\frac{N}{2}$ butterflies of the type as shown in figure 7.9. The number of stages P in the structure shall be $P = \log_2 N$. For 8-point DFT computation i.e. N=8 and hence number of stages, $P = \log_2 N = \log_2 8 = 3$ which is shown in figure 7.10.

Therefore,

$$\text{Speed improvement factor} = \frac{\text{Complex multiplication of direct computation}}{\text{Complex multiplication in FFT algorithm}}$$

$$= \left[\frac{N^2}{\frac{N}{2} \log_2 N} \right] = \frac{2N}{\log_2 N}$$

How efficient is the FFT?

The DFT takes N^2 operations for N points. Since at any stage the computation required to combine smaller DFTs into larger DFTs is proportional to N, and there are $\log_2 N$ stages (for radix 2), the total computation is proportional to $N * \log_2 N$. Therefore, the ratio between a DFT computation and an FFT computation for the same N is proportional to $\frac{N}{\log_2 N}$. In cases where N is small this ratio is not very significant, but when N becomes large, this ratio gets very large. (Every time you double N, the numerator doubles, but the denominator only increases by 1.)